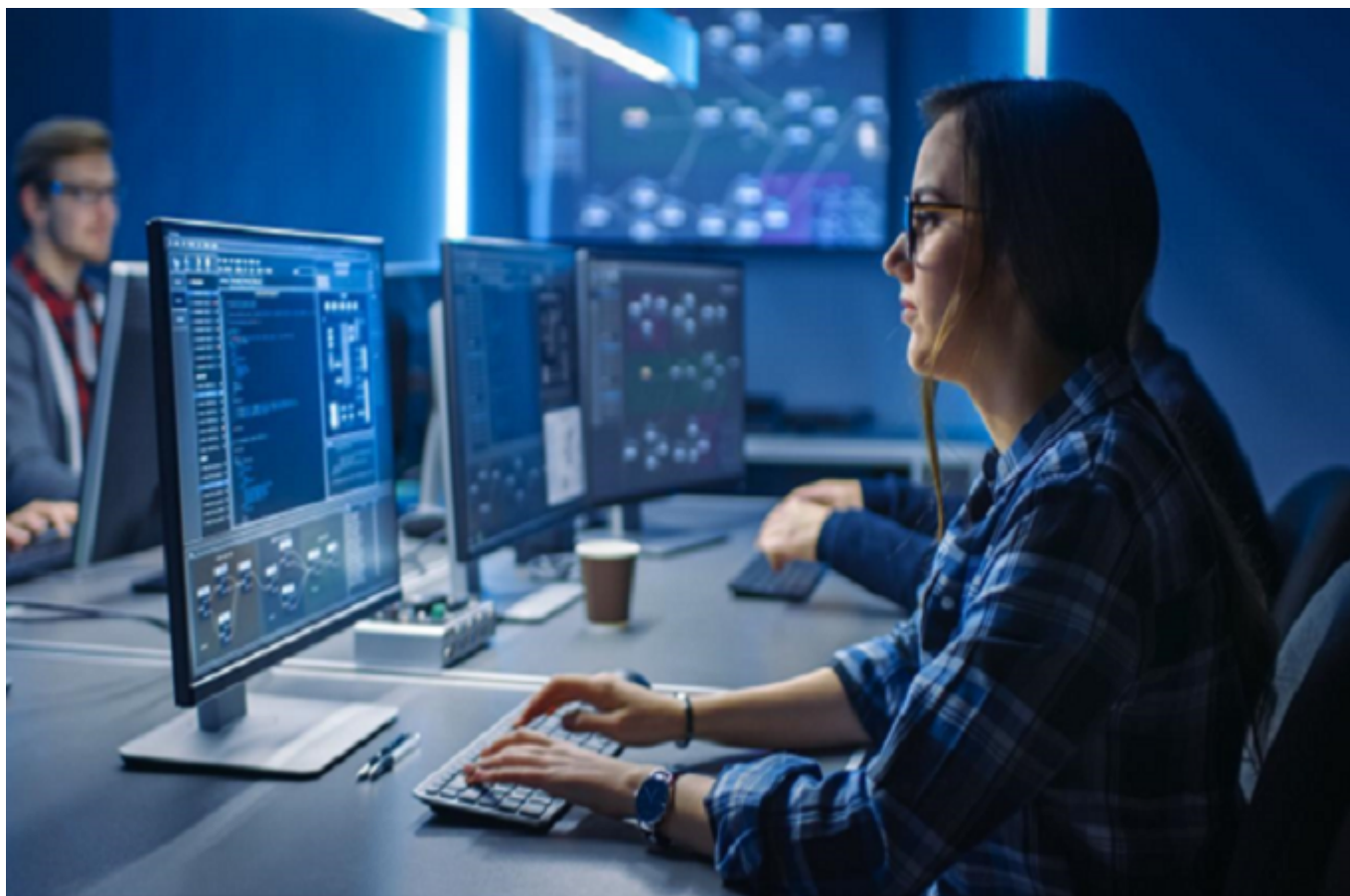




Security Analyst

QP Code: SSC/Q0901

Version: 4.0

NSQF Level: 5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
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Qualification Pack

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Qualification Pack

SSC/Q0901: Security Analyst

Brief Job Description

Individuals at this job are responsible for protecting organizational data by monitoring and analyzing threats, coordinating responses to information security incidents, managing and securing cloud environments, and executing phishing and social engineering defenses. They ensure robust security measures, mitigate risks, and maintain compliance with industry standards across digital platforms.

Personal Attributes

This job may require the individual to work independently and take decisions for his/her own area of work. The individual should be result oriented and have a high attention for detail. The individual should also be able to demonstrate communication skills, logical thinking along with willingness to undertake desk-based job with long hours.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0954: Monitor and Analyze Threats](#)
2. [SSC/N0955: Coordinate with team and respond to Incidents](#)
3. [SSC/N0956: Manage and Secure Cloud Data](#)
4. [SSC/N0957: Execute Phishing and Social Engineering Defense](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Information and Cyber Security
Country	India
NSQF Level	5
Credits	16

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 2522.0201
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (of 3-year/ 4-year UG) with NA of experience OR Completed 3-year diploma (after 10th) with 1.5 years of experience in relevant field OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	4.0
Reference code on NQR	QG-05-IT-03639-2025-V2-NASSCOM
NQR Version	4.0

Qualification Pack

SSC/N0954: Monitor and Analyze Threats

Description

The OS unit is about identifying, analyzing, and responding to potential security threats to ensure the protection and integrity of digital systems.

Scope

The scope covers the following :

- Threat Monitoring and Analysis
- Digital Forensics and Incident Investigation
- Threat Analysis, Incident Investigation, and Reporting

Elements and Performance Criteria

Threat Monitoring and Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** monitor security alerts from firewalls, IDS, SIEM, and other security platforms continuously, ensuring immediate identification of potential threats
- PC2.** perform real-time analysis of security alerts, correlating them with system data to detect suspicious activities
- PC3.** ensure alerts are cross-verified with other logs (e.g., network, application) to confirm the legitimacy of the threat
- PC4.** categorize identified threats based on severity, impact on critical business operations, and sensitivity of data potentially compromised
- PC5.** utilize frameworks like MITRE ATT&CK to classify and understand the tactics, techniques, and procedures (TTPs) used in attacks
- PC6.** apply a risk-based approach to determine the urgency and priority of threat response

Digital Forensics and Incident Investigation

To be competent, the user/individual on the job must be able to:

- PC7.** initiate incident investigations by collecting and preserving digital evidence from compromised systems, network devices, and endpoints following best practices
- PC8.** perform forensic analysis to reconstruct the sequence of events leading to a security breach or attempted attack
- PC9.** analyze collected data to pinpoint the root cause and identify attack vectors, aiding in understanding how the system was compromised
- PC10.** collaborate with the incident response team to isolate affected systems and mitigate ongoing threats
- PC11.** recommend remediation steps based on the investigation, including patching vulnerabilities, hardening systems, or improving security controls
- PC12.** use appropriate security tools (e.g., IDS, firewalls, SIEM) proficiently to detect, analyze, and respond to cyber threats

Threat Analysis, Incident Investigation, and Reporting

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC13.** demonstrate knowledge of network protocols, system vulnerabilities, and attack vectors in the analysis and categorization of threats
- PC14.** analyze complex and voluminous data sets to identify patterns or anomalies indicative of potential security incidents
- PC15.** troubleshoot and resolve issues arising during the detection, analysis, and response phases of incident management
- PC16.** effectively communicate findings and analysis results to both technical and non-technical stakeholders
- PC17.** stay updated on emerging cyber threats, vulnerabilities, and security best practices, and apply this knowledge to improve monitoring and defense strategies
- PC18.** regularly update incident response and analysis techniques based on lessons learned from past incidents and evolving threat landscapes
- PC19.** document the analysis and findings in a detailed incident report for further action and knowledge sharing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles of cybersecurity, including network security, endpoint security, and cloud security, to detect and mitigate threats effectively.
- KU2.** Various types of cyber threats, attack vectors, and vulnerabilities, including malware, phishing, ransomware, and denial-of-service (DoS) attacks.
- KU3.** The role and functionalities of security tools such as SIEM (Security Information and Event Management), IDS/IPS (Intrusion Detection/Prevention Systems), firewalls, and endpoint detection and response (EDR) solutions.
- KU4.** Cyber threat intelligence frameworks, such as MITRE ATT&CK, to classify attack tactics, techniques, and procedures (TTPs).
- KU5.** Incident response methodologies and digital forensic techniques for analyzing security breaches and determining root causes.
- KU6.** Network protocols, system logs, and traffic patterns to identify anomalies and indicators of compromise (IoCs).
- KU7.** Risk assessment methodologies and prioritization techniques based on business impact and sensitivity of compromised data.
- KU8.** Regulatory compliance requirements, such as GDPR, ISO 27001, and NIST frameworks, for secure data handling and threat mitigation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Monitor and analyze real-time security alerts from multiple sources, including SIEM, IDS/IPS, and firewall logs, to detect potential threats.
- GS2.** Apply forensic analysis techniques to reconstruct the sequence of events in a security breach and determine attack vectors.

Qualification Pack

- GS3.** Communicate technical findings and security incidents clearly to both technical teams and non-technical stakeholders.
- GS4.** Implement security measures, such as system hardening, patch management, and configuration changes, based on investigation findings.
- GS5.** Adapt threat detection and response strategies based on evolving cybersecurity trends, attack patterns, and new vulnerabilities.
- GS6.** Maintain detailed documentation of threat analysis, incident response activities, and remediation measures for compliance and future reference.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Threat Monitoring and Analysis</i>	9	16	-	7
PC1. monitor security alerts from firewalls, IDS, SIEM, and other security platforms continuously, ensuring immediate identification of potential threats	2	3	-	1
PC2. perform real-time analysis of security alerts, correlating them with system data to detect suspicious activities	1	3	-	1
PC3. ensure alerts are cross-verified with other logs (e.g., network, application) to confirm the legitimacy of the threat	1	3	-	1
PC4. categorize identified threats based on severity, impact on critical business operations, and sensitivity of data potentially compromised	2	2	-	1
PC5. utilize frameworks like MITRE ATT&CK to classify and understand the tactics, techniques, and procedures (TTPs) used in attacks	1	3	-	2
PC6. apply a risk-based approach to determine the urgency and priority of threat response	2	2	-	1
<i>Digital Forensics and Incident Investigation</i>	9	17	-	6
PC7. initiate incident investigations by collecting and preserving digital evidence from compromised systems, network devices, and endpoints following best practices	2	2	-	1
PC8. perform forensic analysis to reconstruct the sequence of events leading to a security breach or attempted attack	1	3	-	1
PC9. analyze collected data to pinpoint the root cause and identify attack vectors, aiding in understanding how the system was compromised	2	3	-	1
PC10. collaborate with the incident response team to isolate affected systems and mitigate ongoing threats	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recommend remediation steps based on the investigation, including patching vulnerabilities, hardening systems, or improving security controls	1	3	-	1
PC12. use appropriate security tools (e.g., IDS, firewalls, SIEM) proficiently to detect, analyze, and respond to cyber threats	2	3	-	1
<i>Threat Analysis, Incident Investigation, and Reporting</i>	12	17	-	7
PC13. demonstrate knowledge of network protocols, system vulnerabilities, and attack vectors in the analysis and categorization of threats	1	2	-	1
PC14. analyze complex and voluminous data sets to identify patterns or anomalies indicative of potential security incidents	1	3	-	1
PC15. troubleshoot and resolve issues arising during the detection, analysis, and response phases of incident management	2	2	-	1
PC16. effectively communicate findings and analysis results to both technical and non-technical stakeholders	2	3	-	1
PC17. stay updated on emerging cyber threats, vulnerabilities, and security best practices, and apply this knowledge to improve monitoring and defense strategies	2	2	-	1
PC18. regularly update incident response and analysis techniques based on lessons learned from past incidents and evolving threat landscapes	2	3	-	1
PC19. document the analysis and findings in a detailed incident report for further action and knowledge sharing	2	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0954
NOS Name	Monitor and Analyze Threats
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N0955: Coordinate with team and respond to Incidents

Description

The OS unit is about collaborating with team members to address incidents efficiently, ensuring timely resolution and minimal disruption to operations.

Scope

The scope covers the following :

- Incident Escalation and Threat Response Management
- Post-Incident Analysis and Vulnerability Management

Elements and Performance Criteria

Incident Escalation and Threat Response Management

To be competent, the user/individual on the job must be able to:

- PC1.** establish and document clear escalation procedures for major security incidents, ensuring timely notification of Incident Response Team (IRT) and Threat Hunting Team
- PC2.** effectively communicate threat status, urgency, and required actions to relevant teams and management during an incident
- PC3.** isolate the compromised system
- PC4.** implement and configure SOAR tools to automate threat detection, investigation, and containment processes
- PC5.** refer to the organization's knowledge base to review past information security incidents and the methods used to handle them
- PC6.** allocate information security incidents to the relevant personnel for investigation and response
- PC7.** evaluate the severity and potential impact of identified threats, prioritizing them based on organizational risk and criticality
- PC8.** monitor the progress of investigations into information security incidents and escalate to the relevant personnel if the progress fails to meet established standards or service level agreements (SLAs)

Post-Incident Analysis and Vulnerability Management

To be competent, the user/individual on the job must be able to:

- PC9.** conduct thorough post-incident investigations to determine the root cause of the security breach or attempted intrusion
- PC10.** provide input based on security trends and incident analysis to strengthen policy development and enforcement mechanisms
- PC11.** conduct regular vulnerability assessments to identify potential weaknesses in the organization's systems, networks, and applications
- PC12.** coordinate with IT teams to prioritize and apply security patches to address known vulnerabilities in a timely manner

Qualification Pack

- PC13.** effectively analyze complex data sets to detect patterns of suspicious activity or potential security breaches
- PC14.** adhere to the organization's policies, standards, procedures, guidelines, and service level agreements (SLAs) while coordinating responses to information security incidents

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles of incident response, including escalation procedures, threat classification, and communication protocols for security incidents.
- KU2.** The configuration, implementation, and use of Security Orchestration, Automation, and Response (SOAR) tools for threat detection and containment.
- KU3.** Methods for evaluating the severity and potential impact of cybersecurity threats, prioritizing incidents based on business risk, and determining mitigation strategies.
- KU4.** The role of Incident Response Teams (IRT) and Threat Hunting Teams in managing security breaches and coordinating responses.
- KU5.** Techniques for conducting root cause analysis post-incident to determine security vulnerabilities, attack vectors, and defensive improvements.
- KU6.** Best practices for conducting regular vulnerability assessments, identifying weaknesses in networks, systems, and applications, and applying security patches in coordination with IT teams.
- KU7.** Organizational policies, service level agreements (SLAs), and compliance requirements that govern information security incident management and response.
- KU8.** The fundamentals of data analysis in cybersecurity, including how to interpret complex security logs, detect anomalies, and analyze trends in security events.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively communicate security incidents, their urgency, and required actions to technical teams, management, and other stakeholders.
- GS2.** Monitor the progress of ongoing security investigations and escalate unresolved issues to appropriate personnel per organizational standards.
- GS3.** Apply data analysis techniques to detect suspicious activity, analyze attack patterns, and improve detection mechanisms.
- GS4.** Coordinate with IT teams to prioritize and apply security patches, ensuring a timely response to known vulnerabilities.
- GS5.** Document security incidents, including findings from post-incident investigations, in a structured format for reporting and policy development.
- GS6.** Adhere to cybersecurity policies, procedures, and SLAs while managing information security incidents and coordinating responses.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Incident Escalation and Threat Response Management</i>	16	30	-	12
PC1. establish and document clear escalation procedures for major security incidents, ensuring timely notification of Incident Response Team (IRT) and Threat Hunting Team	2	4	-	2
PC2. effectively communicate threat status, urgency, and required actions to relevant teams and management during an incident	2	4	-	1
PC3. isolate the compromised system	-	4	-	1
PC4. implement and configure SOAR tools to automate threat detection, investigation, and containment processes	2	4	-	2
PC5. refer to the organization's knowledge base to review past information security incidents and the methods used to handle them	3	5	-	2
PC6. allocate information security incidents to the relevant personnel for investigation and response	3	3	-	1
PC7. evaluate the severity and potential impact of identified threats, prioritizing them based on organizational risk and criticality	2	3	-	1
PC8. monitor the progress of investigations into information security incidents and escalate to the relevant personnel if the progress fails to meet established standards or service level agreements (SLAs)	2	3	-	2
<i>Post-Incident Analysis and Vulnerability Management</i>	14	20	-	8
PC9. conduct thorough post-incident investigations to determine the root cause of the security breach or attempted intrusion	2	3	-	2
PC10. provide input based on security trends and incident analysis to strengthen policy development and enforcement mechanisms	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. conduct regular vulnerability assessments to identify potential weaknesses in the organization's systems, networks, and applications	3	4	-	1
PC12. coordinate with IT teams to prioritize and apply security patches to address known vulnerabilities in a timely manner	2	4	-	1
PC13. effectively analyze complex data sets to detect patterns of suspicious activity or potential security breaches	3	2	-	2
PC14. adhere to the organization's policies, standards, procedures, guidelines, and service level agreements (SLAs) while coordinating responses to information security incidents	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0955
NOS Name	Coordinate with team and respond to Incidents
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N0956: Manage and Secure Cloud Data

Description

The OS unit is about implementing data security measures and managing cloud storage to ensure the confidentiality, integrity, and availability of digital information.

Scope

The scope covers the following :

- Cloud Security Management and Threat Protection
- Implement and Manage Identity, Access, and Data Protection Policies
- Cloud Security Management and Disaster Recovery

Elements and Performance Criteria

Cloud Security Management and Threat Protection

To be competent, the user/individual on the job must be able to:

- PC1.** configure and manage threat protection, information protection, and identity protection features within the Microsoft 365 Defender Suite
- PC2.** deploy and manage Azure Security Center, Azure Firewall, and Azure Key Vault effectively
- PC3.** regularly review and update security configurations to address emerging threats and vulnerabilities
- PC4.** implement security measures across multiple cloud providers and servers
- PC5.** utilize SIEM tools and threat intelligence feeds to monitor cloud environments continuously
- PC6.** identify and report suspicious activities or anomalies within specified response times
- PC7.** investigate security incidents promptly and efficiently, documenting findings and remediation steps
- PC8.** implement corrective actions to mitigate the impact of incidents and prevent recurrence
- PC9.** manage user accounts, ensuring appropriate access based on roles and responsibilities
- PC10.** maintain records of user access changes and ensure timely de-provisioning of inactive accounts

Implement and Manage Identity, Access, and Data Protection Policies

To be competent, the user/individual on the job must be able to:

- PC11.** implement and manage Multi-Factor Authentication (MFA) for all relevant user accounts, achieving a specified compliance rate
- PC12.** conduct regular access reviews, identifying and revoking unnecessary user permissions
- PC13.** identify and classify sensitive data types (e.g., PII, PHI) effectively within cloud environments
- PC14.** categorize data according to sensitivity levels and ensure appropriate protection measures are in place
- PC15.** implement and monitor DLP policies to prevent unauthorized data access and transmission
- PC16.** ensure adherence to industry regulations (e.g., GDPR, HIPAA, PCI DSS) by conducting regular audits and assessments

Qualification Pack

PC17. develop and deliver security awareness training programs for employees, achieving a specified participation rate

Cloud Security Management and Disaster Recovery

To be competent, the user/individual on the job must be able to:

PC18. conduct regular vulnerability assessments on cloud environments, documenting findings and remediation efforts

PC19. ensure timely application of security patches and updates across all cloud services and applications

PC20. develop and implement backup procedures to ensure data integrity and availability

PC21. participate in disaster recovery drills, documenting lessons learned and updating recovery plans accordingly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The principles of threat protection, information security, and identity management within the Microsoft 365 Defender Suite, including key components such as Microsoft Defender for Endpoint, Defender for Office 365, and Defender for Identity.

KU2. The configuration, deployment, and management of Azure Security Center, Azure Firewall, and Azure Key Vault to ensure secure cloud infrastructure and compliance with security best practices.

KU3. The methodologies for continuously monitoring security configurations and updating them to address new threats, vulnerabilities, and regulatory changes.

KU4. Strategies for implementing multi-cloud security measures, ensuring robust security controls across different cloud service providers and on-premises infrastructure.

KU5. The functions and capabilities of Security Information and Event Management (SIEM) tools and threat intelligence feeds in detecting, analyzing, and responding to potential security threats in real time.

KU6. Incident response processes, including identifying, documenting, and investigating suspicious activities, anomalies, and security breaches while maintaining compliance with security policies.

KU7. Identity and access management principles, including user provisioning, de-provisioning, access reviews, and the implementation of Multi-Factor Authentication (MFA) to maintain security compliance.

KU8. The classification of sensitive data, including Personally Identifiable Information (PII) and Protected Health Information (PHI), and the implementation of Data Loss Prevention (DLP) policies to prevent unauthorized data exposure.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Effectively communicate security incidents, response actions, and remediation plans to technical teams, management, and other stakeholders.

Qualification Pack

- GS2.** Analyze and interpret security logs, identify attack patterns, and use threat intelligence insights to enhance security monitoring and detection mechanisms.
- GS3.** Coordinate with IT teams and compliance officers to ensure adherence to security standards, including GDPR, HIPAA, and PCI DSS, by conducting audits and assessments.
- GS4.** Perform root cause analysis on security incidents to determine vulnerabilities, document findings, and implement corrective actions to prevent recurrence.
- GS5.** Develop and conduct security awareness training programs for employees to enhance their understanding of cybersecurity risks, policies, and best practices.
- GS6.** Organize and participate in disaster recovery drills, ensuring that backup procedures are tested, recovery plans are updated, and lessons learned are documented.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cloud Security Management and Threat Protection</i>	14	26	-	9
PC1. configure and manage threat protection, information protection, and identity protection features within the Microsoft 365 Defender Suite	1	3	-	1
PC2. deploy and manage Azure Security Center, Azure Firewall, and Azure Key Vault effectively	1	2	-	1
PC3. regularly review and update security configurations to address emerging threats and vulnerabilities	1	3	-	1
PC4. implement security measures across multiple cloud providers and servers	2	3	-	1
PC5. utilize SIEM tools and threat intelligence feeds to monitor cloud environments continuously	1	2	-	1
PC6. identify and report suspicious activities or anomalies within specified response times	1	3	-	1
PC7. investigate security incidents promptly and efficiently, documenting findings and remediation steps	2	2	-	1
PC8. implement corrective actions to mitigate the impact of incidents and prevent recurrence	1	2	-	-
PC9. manage user accounts, ensuring appropriate access based on roles and responsibilities	2	3	-	1
PC10. maintain records of user access changes and ensure timely de-provisioning of inactive accounts	2	3	-	1
<i>Implement and Manage Identity, Access, and Data Protection Policies</i>	10	15	-	7
PC11. implement and manage Multi-Factor Authentication (MFA) for all relevant user accounts, achieving a specified compliance rate	1	3	-	1
PC12. conduct regular access reviews, identifying and revoking unnecessary user permissions	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify and classify sensitive data types (e.g., PII, PHI) effectively within cloud environments	2	2	-	1
PC14. categorize data according to sensitivity levels and ensure appropriate protection measures are in place	1	2	-	1
PC15. implement and monitor DLP policies to prevent unauthorized data access and transmission	1	2	-	1
PC16. ensure adherence to industry regulations (e.g., GDPR, HIPAA, PCI DSS) by conducting regular audits and assessments	3	2	-	1
PC17. develop and deliver security awareness training programs for employees, achieving a specified participation rate	1	2	-	1
<i>Cloud Security Management and Disaster Recovery</i>	6	9	-	4
PC18. conduct regular vulnerability assessments on cloud environments, documenting findings and remediation efforts	1	2	-	1
PC19. ensure timely application of security patches and updates across all cloud services and applications	2	2	-	1
PC20. develop and implement backup procedures to ensure data integrity and availability	1	3	-	1
PC21. participate in disaster recovery drills, documenting lessons learned and updating recovery plans accordingly	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0956
NOS Name	Manage and Secure Cloud Data
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

SSC/N0957: Execute Phishing and Social Engineering Defense

Description

The OS unit is about identifying, preventing, and responding to phishing and social engineering attacks to safeguard sensitive information and maintain cyber security.

Scope

The scope covers the following :

- Phishing Threat Analysis and Response Management
- Phishing Threat Management and Incident Response

Elements and Performance Criteria

Phishing Threat Analysis and Response Management

To be competent, the user/individual on the job must be able to:

- PC1.** employ advanced tools (e.g., Phishme) to analyze and assess the risk level of incoming emails
- PC2.** identify phishing attempts by examining emails for red flags such as suspicious sender addresses, urgent requests, and grammatical errors
- PC3.** conduct thorough risk assessments for potential phishing threats, factoring in data sensitivity and employee vulnerability
- PC4.** design and implement phishing simulation exercises to educate employees about phishing tactics
- PC5.** analyze employee responses to phishing simulations to identify knowledge gaps and areas needing improvement
- PC6.** employ digital forensics techniques to trace the origin of suspicious emails, utilizing IP addresses, DNS records, and metadata
- PC7.** investigate incidents of successful phishing attacks to assess damage and identify corrective actions to prevent recurrence
- PC8.** develop, implement, and update phishing prevention policies, ensuring compliance with industry standards and regulations
- PC9.** actively contribute to the development and maintenance of incident response plans related to phishing and social engineering attacks
- PC10.** ensure readiness for a coordinated response to phishing incidents, involving relevant stakeholders and utilizing established protocols
- PC11.** ensure compliance with data privacy and security regulations, conducting regular audits and assessments as necessary
- PC12.** prepare and present reports on phishing threat trends, employee training effectiveness, and incident response outcomes to management

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** The principles of phishing threat management, including common attack techniques, indicators of phishing attempts, and methods to identify suspicious emails.
- KU2.** The configuration, implementation, and use of phishing detection tools (e.g., PhishMe) for analyzing incoming emails, assessing risk levels, and identifying potential threats.
- KU3.** Techniques for conducting phishing risk assessments by evaluating data sensitivity, employee susceptibility, and overall organizational risk exposure.
- KU4.** Best practices for designing and executing phishing simulation exercises to test employee awareness and improve security training programs.
- KU5.** Digital forensics techniques for investigating phishing incidents, including analyzing email headers, tracing IP addresses, and examining DNS records.
- KU6.** Incident response strategies for handling phishing attacks, including response planning, stakeholder coordination, and mitigation procedures.
- KU7.** The regulatory and compliance requirements related to data privacy and security, including industry standards such as GDPR, HIPAA, and ISO 27001.
- KU8.** Methods for preparing, analyzing, and presenting phishing threat intelligence reports, including tracking trends, evaluating training effectiveness, and recommending security improvements.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively analyze emails for phishing indicators such as suspicious sender addresses, urgent language, and grammatical inconsistencies.
- GS2.** Implement and manage phishing prevention policies, ensuring compliance with industry standards and organizational security frameworks.
- GS3.** Conduct phishing awareness training sessions and simulations, assessing employee responses and identifying knowledge gaps.
- GS4.** Investigate phishing incidents, document findings, and recommend corrective actions to prevent future attacks.
- GS5.** Communicate phishing threats, response strategies, and security recommendations to technical teams, management, and employees.
- GS6.** Collaborate with security teams to ensure compliance with data privacy regulations, conducting regular audits and assessments to maintain security integrity.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Phishing Threat Analysis and Response Management</i>	30	50	-	20
PC1. employ advanced tools (e.g., Phishme) to analyze and assess the risk level of incoming emails	2	4	-	2
PC2. identify phishing attempts by examining emails for red flags such as suspicious sender addresses, urgent requests, and grammatical errors	2	4	-	2
PC3. conduct thorough risk assessments for potential phishing threats, factoring in data sensitivity and employee vulnerability	3	5	-	2
PC4. design and implement phishing simulation exercises to educate employees about phishing tactics	3	6	-	2
PC5. analyze employee responses to phishing simulations to identify knowledge gaps and areas needing improvement	3	4	-	1
PC6. employ digital forensics techniques to trace the origin of suspicious emails, utilizing IP addresses, DNS records, and metadata	3	4	-	1
PC7. investigate incidents of successful phishing attacks to assess damage and identify corrective actions to prevent recurrence	2	5	-	2
PC8. develop, implement, and update phishing prevention policies, ensuring compliance with industry standards and regulations	3	4	-	2
PC9. actively contribute to the development and maintenance of incident response plans related to phishing and social engineering attacks	2	4	-	1
PC10. ensure readiness for a coordinated response to phishing incidents, involving relevant stakeholders and utilizing established protocols	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure compliance with data privacy and security regulations, conducting regular audits and assessments as necessary	2	3	-	2
PC12. prepare and present reports on phishing threat trends, employee training effectiveness, and incident response outcomes to management	3	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0957
NOS Name	Execute Phishing and Social Engineering Defense
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
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Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.
7. Candidate needs to achieve 70% or more for successfully passing the QP.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0954.Monitor and Analyze Threats	30	50	0	20	100	22
SSC/N0955.Coordinate with team and respond to Incidents	30	50	0	20	100	22
SSC/N0956.Manage and Secure Cloud Data	30	50	0	20	100	22
SSC/N0957.Execute Phishing and Social Engineering Defense	30	50	0	20	100	22
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	12
Total	140	230	-	80	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.